HO CHI MINH CITY – UNIVERSITY OF TECHNOLOGY

LAB 4 REPORT

THERMOCOUPLE
DETERMINATION OF THE THERMOCOUPLE CONSTANT

Instructor: Dr. Tran Trung Nghia
Subject: General Physics Labs
Class: CC02
School year: 2021-2022 - Semester 1

BACHKHOACNCP.COM
TÀI LIỆU SƯU TẬP
BỞI HCMUT-CNCP

GROUP 1 MEMBERS

Students' names	ID code	Lecturer's comment
1.		
2.		
3.	2	
4.	2	
5.		

I/ OBJECTIVE

To get used to thermocouple (chromel-alumel), MC-897 electronic millivoltmeter to determine the thermocouple constant.

II/ THEORY

A thermocouple is a closed electric circuit consisting of two dissimilar conductors A and B that are joined together (Figure 1). If junctions 1 and 2 are at different temperatures T1 and T2 (T1>T2), respectively, an electric current that is called *thermoelectric current* is produced within the circuit. Electromotive force that generates the thermoelectric current is known as *thermal electromotive force E*.

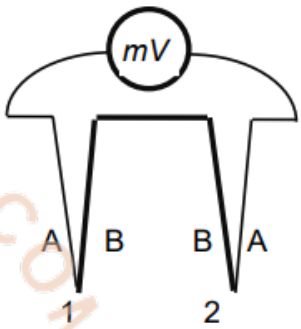


Figure 1

Thermal electromotive force is yielded because free electrons diffuse from the metal with a higher density of free electrons to the metal with a lower density at junctions 1 and 2. Theories and experiments have proved that the magnitude of thermal electromotive force depends on the nature of the two metals and the temperature difference between the two junctions of the thermocouple. Within a certain temperature range (depending on thermocouple types), thermal electromotive force E increases proportionally to the temperature difference (T1 - T2) between the two junctions as denoted in Equation 1:

$$E_{nđ} = C.(T1 - T2) \qquad (1)$$

where C is the thermocouple constant. For metal thermocouples, the value of E is small; therefore, C is measured in the unit of either $\frac{\mu V}{V}$ or $\frac{\mu V}{^{\circ}C}$. Thermocouples can be used as a thermometer to measure temperature or a low-power electric source (obtained by converting thermal energy to electric energy).

Prior the use of a thermocouple, we have to sample the thermocouple; that is, we have to investigate the

temperature range (depending on the thermocouple type) where thermal electromotive force is linearly dependent on the temperature difference (T1 - T2) between its two junctions. As a result, the graph representing the function of $E = C.(T1 - T2)$ is a segment (Figure 2), and constant C is calculated as in Equation 2:

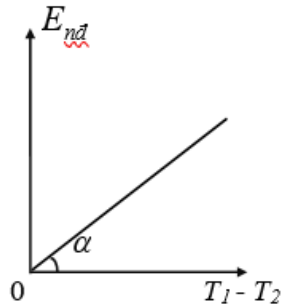


Figure 2

$$C = \frac{\square\square đ}{\square 1 - \square 2} = tga \quad (2)$$

Since resistance RV of millivoltmeter **mV** is much larger than internal resistance *r* of the thermocouple, applying Ohm's law to a closed circuit, we have:

$$E = I.(R_V + r) \approx I.R_V = U_V \quad (3)$$

where *I* is the current intensity, *Uv* is the reading on millivoltmeter **mV**.

Thus, for every value of temperature difference *(T1 - T2)* between the hot junction (T1) and the cold junction (T2) of the thermocouple, we measure the corresponding value of thermal electromotive force E on millivoltmeter **mV**. Therefore, we can draw the graph representing thermocouple sampling and determine the value of constant C.

BACHKHOACNCP.COM
TÀI LIỆU SƯU TẬP
BỞI HCMUT-CNCP

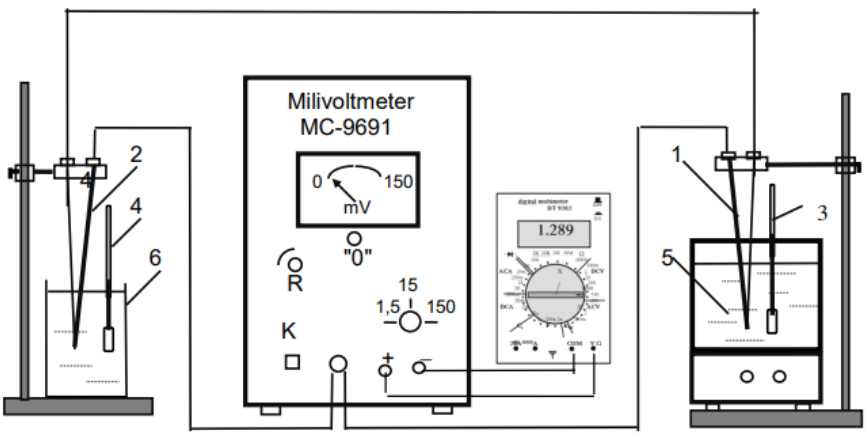


Figure 3

III/ APPARATUS

1. Thermocouple (chromel-alumel);
2. Thermocouple holder;
3. MC-897 electronic millivoltmeter;
4. Two 250ml experiment glasses;
5.Digital thermometer(-500C □1500C);
6. Wine thermometer (0 □ 1000C);
7. Boiler (220V - 300W);
8. Digital multimeter 3,1/2 Digits
9. Set of wires.

IV/ PROCEDURE

Step 1

Connect the wires to form the electric circuit as demonstrated in Figure 3. Set the range selector knob of millivoltmeter **mV** at position 15. Set the digital multimeter at the measurement range of 200mVDC. Use a piece of wire to short circuit the positive and negative probes. Turn sensitivity knob **R** counterclockwise all the way left.

Plug digital millivoltmeter **mV** in the mains of 220V. Press button K on the front and millivoltmeter **mV** starts. Wait for 4 ÷ 5 minutes, then perform zeroing the measurement range of millivoltmeter **mV** by turning slowly the zero corrector to move the pointer to 0, and the digital multimeter also displays 0. Then, the maximum value of the measurement range on the millivoltmeter mV is equal to 15*mV*. Turn knob "0" until the reading is 5mV on the electronic millivoltmeter, then read the value of $U_{kđ}$ on the digital multimeter to derive the amplification coefficient k = $U_{kđ}$ /5mV. Afterward, repeat zeroing for millivoltmeter **mV**, disconnect the wire from the positive and negative probes and connect it to the thermocouple.

Step 2

Place the bulbs of thermometers **3** and **4** at the same level as the junctions of thermocouples **1** and **2** in the middle of boiler **5** and experimental glass **6**, respectively. Wait for 4÷5 minutes, then read and record the reading of temperature T_1 on thermometer **3** and that of temperature T_2 on thermometer **4** in Table 1. If T_1 is equal to T_2, the pointer of millivoltmeter **mV** must point at 0.

BACHKHOACNCP.COM TÀI LIỆU SƯU TẬP BỞI HCMUT-CNCP

Step 3

Plug the boiler **5** in the mains of 220V. Wait until the water in the boiler begins to boil, then unplug the boiler **5** and let the water in the boiler cool down (avoid a quick cool down).

For every value of T_1, record the reading of $U_{kđi}$ on the digital multimeter to calculate thermal electromotive force $E_{nđ}$ and write it down in Table 1. Repeat this process until temperature T_1 decreases to 40°C.

***Note*:** When the range selector "RANGE" of millivoltmeter **mV** is set at X10 and sensitivity knob R is turned all the way left, the maximum limit on millivoltmeter **mV** is 15mV.

Step 4

Read and record the following data in Table 1:

a. The level of accuracy δ_v and the maximum value U_m on the measurement range of electronic millivoltmeter **mV**.

b. The level of accuracy δ_v and the maximum value U_m on the measurement range of digital millivoltmeter **mV**.

c. Accuracy ΔT of thermometers **3** and **4**.

V/ EQUATIONS

1. **Determine amplification factor k and relative error**

$$k = \frac{U_h}{U_c} \qquad \frac{\Delta k}{k} = \frac{\Delta U_h}{U_h} + \frac{\Delta U_c}{U_c}$$

2. **Calculate absolute error ΔU_c and ΔU_h**

$$\Delta U_k = \delta_V . U_{max}\ (mV) \qquad \Delta U_h = U_c . U_h + U\ (U)$$

3. **Determine thermodynamic capacity at i^{th} measurement and the corresponding relative error**

$$C_{nhiệt} = \frac{Q_{đi}}{T}$$

$$\frac{\Delta E_{nđ\,i}}{E_{nđ\,i}} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\,i}}{U_{kđ\,i}} = \varepsilon_i$$

4. **Calculate absolute error $\Delta E_{nđ}$ at i^{th} measurement**

$\Delta E_{nđ\,i} = \varepsilon_i . E_{nđ\,i}\ (mV)$

5. **Calculate absolute error** $\Delta(T_1 - T_2)$

$\Delta(\square_1 - \square_2) = \square\square_1 + \square\square_2 = 2\square\square\ (\square\square)$

6. **Calculate the average** $\underline{C}$ **and absolute error** ΔC

$$\underline{C} = \frac{\sum_{i=1}^{n} (T_{1\,i} - T_2).E_{nđ\,i}}{\sum_{i=1}^{n} (T_{1\,i} - T_2)^2} \qquad (\Delta C)^2 = \frac{S^2}{\sum_{i=1}^{n} (T_{1\,i} - T_2)^2}$$

$$\square^2 = \frac{1}{(n-1)} \sum_{i=1}^{n} (E_{nđ\,i} - \square_{nđ\,i}) \text{ where } \square_{nđ\,i} = C.(T_{1\,i} - T_2)$$

VI/ EXPERIMENTAL DATA

Table 1	- Electronic millivoltmeter **mV**: δ_v = 1,5% ; U_m =15 (mV) - Digital millivoltmeter **mV**: δ_v = 0,5% ; n =3, a=0.1 - Digital thermometer: ΔT =0.3 (°C) - Alcohol thermometer ΔT =1 (°C)					
Measurement/ T_2 =28 (°C)	T_1 (°C)	$T_1 - T_2$	$\Delta(T_1 - T_2)$	$U_{kđ}$ (mV)	$E_{nđ}$ (mV)	$\Delta E_{nđ}$
4	80	52	1,3	127	4,2333	*0,2513*
5	75	47	1,3	119	3,9667	*0,2361*
6	70	42	1,3	103	3,4333	*0,2057*
7	65	37	1,3	95	3,1667	0,1905
8	60	32	1,3	83	2,7667	*0,1677*
9	55	27	1,3	70	2,3333	*0,1430*
10	50	22	1,3	58	1,9333	0,1202

VII/ CALCULATIONS

1. **Calculate values and errors:**

Calculate k and errors:

$\square = \frac{\square_{h\square}}{\square_{\square}} = \frac{150}{5} = 30$

$$\frac{\Delta k}{k} = \frac{\Delta U_{hs}}{U_{hs}} + \frac{\Delta U_k}{U_k} = \frac{1{,}05}{150} + \frac{0{,}225}{5} = 0{,}052$$

$$\Delta k = 0{,}052\,.k = 0{,}052\,.30 = 1{,}56$$

$$\Delta U_k = \delta_V . U_{max} = 1{,}5\% . 15 = 0{,}225\ (mV)$$

$$\Delta \square_{h\square} = \square_{\square} . \square_{h\square} + \square\square = \boldsymbol{0{,}5\% . 150 + 3 . 0{,}1} = \boldsymbol{1{,}05}\ (\square\square)$$

$$\Delta(\square_1 - \square_2) = \square\square_1 + \square\square_2 = \boldsymbol{1 + 0{,}3 = 1{,}3}\ (°C)$$

Calculate $\square_{\square\square}$ at i^{th} measurement

$$\square_{\square\square\,4} = \frac{\square_{\square đ\,4}}{\square} = \frac{127}{30} = 4{,}2333\ (mV)$$

$$\square_{\square\square\,5} = \frac{\square_{\square đ\,5}}{\square} = \frac{119}{30} = 3{,}9667\ (mV)$$

$$\square_{\square\square\,6} = \frac{\square_{\square đ\,6}}{\square} = \frac{103}{30} = 3{,}4333\ (mV)$$

$$\square_{\square\square\,7} = \frac{\square_{\square đ\,7}}{\square} = \frac{95}{30} = 3{,}1667\ (mV)$$

$$\square_{\square\square\,8} = \frac{\square_{\square đ\,8}}{\square} = \frac{83}{30} = 2{,}7667\ (mV)$$

$$\square_{\square\square\,9} = \frac{\square_{\square đ\,9}}{\square} = \frac{70}{30} = 2{,}3333\ (mV)$$

$$\square_{\square\square\,10} = \frac{\square_{\square đ\,10}}{\square} = \frac{58}{30} = 1{,}9333\ (mV)$$

Calculate the corresponding absolute error $\Delta E_{nđ}$ at i^{th} measurement

$$\varepsilon_4 = \frac{\Delta E_{nđ\,4}}{E_{nđ\,4}} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\,4}}{U_{kđ\,4}} = \frac{\Delta k}{k} + \frac{\delta_v \times U_{kđ4} + n \times \alpha}{U_{kđ\,4}}$$

$$= \frac{1{,}56}{30} + \frac{0{,}5\% . 127 + 3 . 0{,}1}{127} = 0{,}0594$$

$$\Delta E_{nđ\,4} = \varepsilon_4 . E_{nđ\,4} = \left(\frac{1{,}56}{30} + \frac{\boldsymbol{0{,}5\% . 127 + 3 . 0{,}1}}{127}\right) . \frac{127}{30} = 0{,}2513\ (mV)$$

$$\varepsilon_5 = \frac{\Delta E_{nđ\,5}}{E_{nđ\,5}} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\,5}}{U_{kđ\,5}} = \frac{\Delta k}{k} + \frac{\delta_v \times U_{kđ5} + n \times \alpha}{U_{kđ\,5}}$$

$$= \frac{1{,}56}{30} + \frac{0{,}5\% . 119 + 3 . 0{,}1}{119} = 0{,}0595$$

$$\Delta E_{nđ\,5} = \varepsilon_5 . E_{nđ\,5} = \left(\frac{1{,}56}{30} + \frac{\boldsymbol{0{,}5\% . 119 + 3 . 0{,}1}}{119}\right) . \frac{119}{30} = 0{,}2361\ (mV)$$

$$\varepsilon_6 = \frac{\Delta E_{nđ\,6}}{E_{nđ\,6}} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\,6}}{U_{kđ\,6}} = \frac{\Delta k}{k} + \frac{\delta_v \times U_{kđ6} + n \times \alpha}{U_{kđ\,6}}$$

$$= \frac{1{,}56}{30} + \frac{0{,}5\% . 103 + 3 . 0{,}1}{103} = 0{,}0599$$

BACHKHOACNCP.COM TÀI LIỆU SƯU TẬP BỞI HCMUT-CNCP

$$\Delta E_{nđ\,6} = \varepsilon_6 \,.\, E_{nđ\,6} = \left(\frac{1{,}56}{30} + \frac{0{,}5\% \,.\, 103 + 3 \,.\, 0{,}1}{103}\right) . \frac{103}{30} = 0{,}2057\ (mV)$$

$$\varepsilon_7 = \frac{\Delta E_{nđ\,7}}{E_{nđ\,7}} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\,7}}{U_{kđ\,7}} = \frac{\Delta k}{k} + \frac{\delta_v \times U_{kđ7} + n \times \alpha}{U_{kđ\,7}}$$

$$= \frac{1{,}56}{30} + \frac{0{,}5\% \,.\, 95 + 3 \,.\, 0{,}1}{95} = 0{,}0602$$

$$\Delta E_{nđ\,7} = \varepsilon_7 \,.\, E_{nđ\,7} = \left(\frac{1{,}56}{30} + \frac{0{,}5\% \,.\, 95 + 3 \,.\, 0{,}1}{95}\right) . \frac{95}{30} = 0{,}1905\ (mV)$$

$$\varepsilon_8 = \frac{\Delta E_{nđ\,8}}{E_{nđ\,8}} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\,8}}{U_{kđ\,8}} = \frac{\Delta k}{k} + \frac{\delta_v \times U_{kđ8} + n \times \alpha}{U_{kđ\,8}}$$

$$= \frac{1{,}56}{30} + \frac{0{,}5\% \,.\, 83 + 3 \,.\, 0{,}1}{83} = 0{,}0606$$

$$\Delta E_{nđ\,8} = \varepsilon_8 \,.\, E_{nđ\,8} = \left(\frac{1{,}56}{30} + \frac{0{,}5\% \,.\, 83 + 3 \,.\, 0{,}1}{83}\right) . \frac{83}{30} = 0{,}1677\ (mV)$$

$$\varepsilon_9 = \frac{\Delta E_{nđ\,9}}{E_{nđ\,9}} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\,9}}{U_{kđ\,9}} = \frac{\Delta k}{k} + \frac{\delta_v \times U_{kđ9} + n \times \alpha}{U_{kđ\,9}}$$

$$= \frac{1{,}56}{30} + \frac{0{,}5\% \,.\, 70 + 3 \,.\, 0{,}1}{70} = 0{,}0613$$

$$\Delta E_{nđ\,9} = \varepsilon_9 \,.\, E_{nđ\,9} = \left(\frac{1{,}56}{30} + \frac{0{,}5\% \,.\, 70 + 3 \,.\, 0{,}1}{70}\right) . \frac{70}{30} = 0{,}1430\ (mV)$$

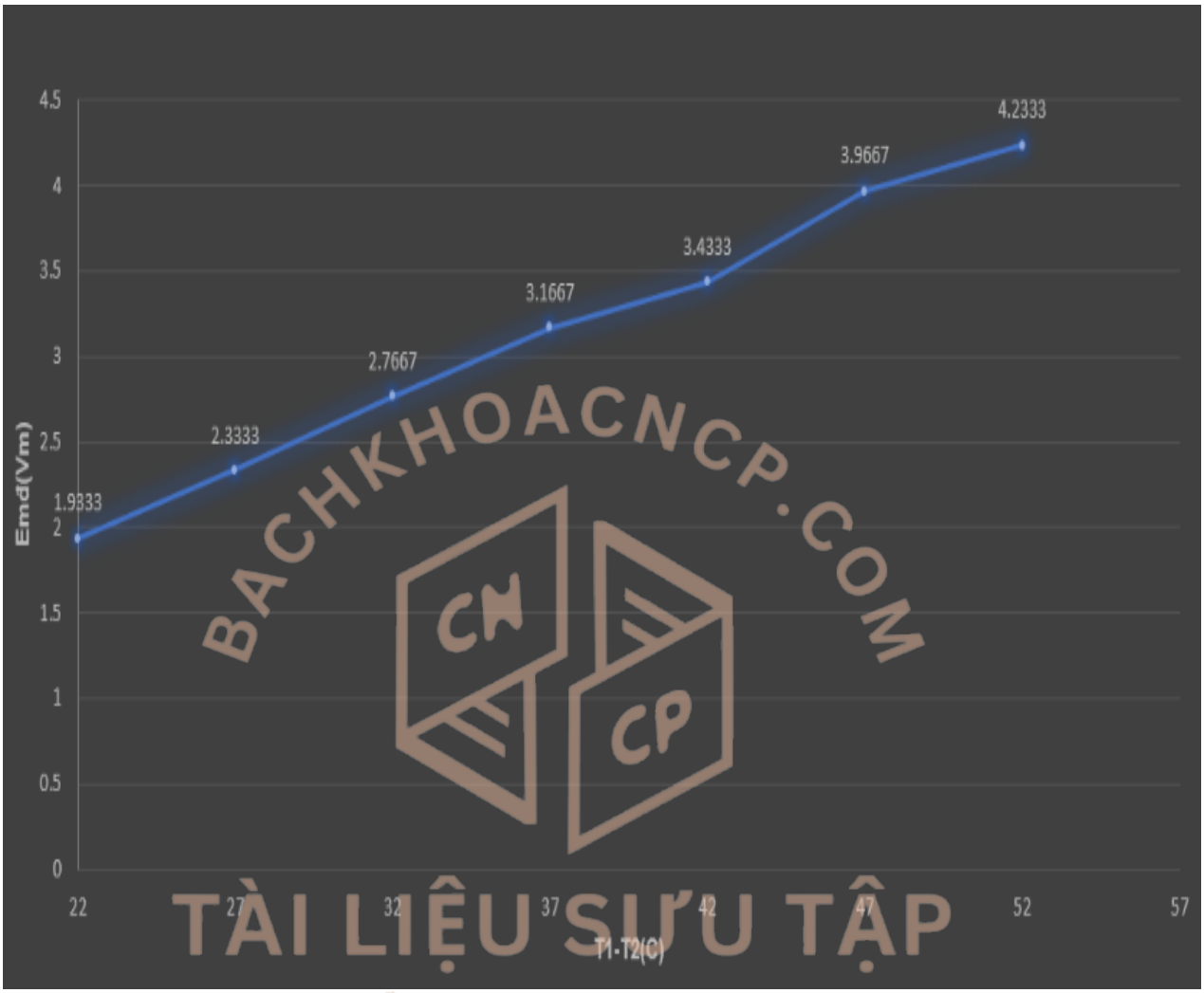
$$\varepsilon_{10} = \frac{\Delta E_{nđ\,10}}{E_{nđ\,10}} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\,10}}{U_{kđ\,10}} = \frac{\Delta k}{k} + \frac{\delta_v \times U_{kđ10} + n \times \alpha}{U_{kđ\,10}}$$

$$= \frac{1{,}56}{30} + \frac{0{,}5\% \,.\, 58 + 3 \,.\, 0{,}1}{58} = 0{,}0622$$

$$\Delta E_{nđ\,10} = \varepsilon_{10} \,.\, E_{nđ\,10} = \left(\frac{1{,}56}{30} + \frac{0{,}5\% \,.\, 58 + 3 \,.\, 0{,}1}{58}\right) . \frac{58}{30} = 0{,}1202\ (mV)$$

2. Plot the graph of Enđ = C.(T1 – T2) on a millimeter paper

Scale: horizontal axis: 5°C/cm; vertical axis: 0,5 mV/cm



BACHKHOACNCP.COM
TÀI LIỆU SƯU TẬP
BỞI HCMUT-CNCP

3. Determination of the thermocouple constant

3.1 Method 1

In the working range of temperature, let the graph representing the function $E = C.T$ be segment OA. Then, the average $\underline{C}$ and the absolute error ΔC of constant C are calculated as follows:

$$\underline{C} = \frac{\sum_{i=1}^{n} (T_{1\,i} - T_2).E_{nđ\,i}}{\sum_{i=1}^{n} (T_{1\,i} - T_2)^2} \qquad (\Delta C)^2 = \frac{S^2}{\sum_{i=1}^{n} (T_{1\,i} - T_2)^2}$$

$$\square^2 = \frac{1}{(n-1)}\sum_{i=1}^{n} (E_{nđ\,i} - \square_{nđ\,i}) \text{ where } \square_{nđ\,i} = C.(T_{1\,i} - T_2)$$

$\underline{C} = \frac{\Sigma}{k}$ or $\xi_{nđi}$ is a point on the line corresponding to $(T_{1i} - T_2)$.

Calculate the average $\underline{C}$ and absolute error ΔC

- $\underline{C} = \frac{\sum_{i=1}^{n} \blacksquare (T_{1\,i} - T_2).E_{nđ\,i}}{\sum_{i=1}^{n} \blacksquare (T_{1\,i} - T_2)^2}$

$$(T_{1\,4} - T_2).E_{nđ\,4} = 52.\frac{127}{30} = 220{,}1333$$

$$(T_{1\,5} - T_2).E_{nđ\,5} = 47.\frac{119}{30} = 186{,}4333$$

$$(T_{1\,6} - T_2).E_{nđ\,6} = 42.\frac{103}{30} = 144{,}2$$

$$(T_{1\,7} - T_2).E_{nđ\,7} = 37.\frac{95}{30} = 117{,}1667$$

$$(T_{1\,8} - T_2).E_{nđ\,8} = 32.\frac{83}{30} = 88{,}5333$$

$$(T_{1\,9} - T_2).E_{nđ\,9} = 27.\frac{70}{30} = 63$$

$$(T_{1\,10} - T_2).E_{nđ\,10} = 22.\frac{58}{30} = 42{,}53333$$

$(T_{1\,4} - T_2)^2 = 52^2$ $(T_{1\,7} - T_2)^2 = 37^2$ $(T_{1\,10} - T_2)^2 = 22^2$

$(T_{1\,5} - T_2)^2 = 47^2$ $(T_{1\,8} - T_2)^2 = 32^2$

$(T_{1\,6} - T_2)^2 = 42^2$ $(T_{1\,9} - T_2)^2 = 27^2$

$$\Rightarrow \sum_{i=4}^{9} \blacksquare (T_{1\,i} - T_2)^2 = 52^2 + 47^2 + 42^2 + 37^2 + 32^2 + 27^2 + 22^2 = 10283$$

$$\sum_{i=4}^{9} \blacksquare (T_{1\,i} - T_2).E_{nđ\,i}$$

$$= 220{,}1333 + 186{,}4333 + 144{,}2 + 117{,}1667 + 88{,}5333 + 63 + 42{,}53333$$

$$=862$$

BACHKHOACNCP.COM TÀI LIỆU SƯU TẬP BỞI HCMUT-CNCP

$$\Rightarrow \underline{C} = \frac{\sum_{i=4}^{9} (T_{1\,i} - T_2).E_{nđ\,i}}{\sum_{i=4}^{9} (T_{1\,i} - T_2)^2} = \frac{862}{10283} \approx 0{,}0838\ (mV/°C)$$

- $\square_{nđ\,i} = C.(T_{1\,i} - T_2)$

$$\square_{nđ\,4} = \underline{C}.(T_{1\,4} - T_2) = \frac{862}{10283}.52 = 4{,}3590\square_{nđ\,5}$$

$$= \underline{C}.(T_{1\,5} - T_2) = \frac{862}{10283}.47 = 3{,}9399\square_{nđ\,6}$$

$$= \underline{C}.(T_{1\,6} - T_2) = \frac{862}{10283}.42 = 3{,}5208\square_{nđ\,7}$$

$$= \underline{C}.(T_{1\,7} - T_2) = \frac{862}{10283}.37 = 3{,}1016\square_{nđ\,8}$$

$$= \underline{C}.(T_{1\,8} - T_2) = \frac{862}{10283}.32 = 2{,}6825$$

$$\square_{nđ\,9} = \underline{C}.(T_{1\,9} - T_2) = \frac{862}{10283}.27 = 2{,}2633$$

BACHKHOACNCP.COM
TÀI LIỆU SƯU TẬP
BỞI HCMUT-CNCP

- $\square^2 = \frac{1}{(n-1)}\sum_{i=1}^{n}\blacksquare(E_{nđ\,i} - \square_{nđ\,i})^2$

$(E_{nđ\,4} - \square_{nđ\,4})^2 = (\frac{127}{30} - \frac{862}{10283}.52)^2 = 0{,}0158$

$(E_{nđ\,5} - \square_{nđ\,5})^2 = (\frac{119}{30} - \frac{862}{10283}.47)^2 = 0{,}00072$

$(E_{nđ\,6} - \square_{nđ\,6})^2 = (\frac{103}{30} - \frac{862}{10283}.42)^2 = 0{,}00764$

$(E_{nđ\,7} - \square_{nđ\,7})^2 = (\frac{95}{30} - \frac{862}{10283}.37)^2 = 0{,}00264$

$(E_{nđ\,8} - \square_{nđ\,8})^2 = (\frac{83}{30} - \frac{862}{10283}.32)^2 = 0{,}00709$

$(E_{nđ\,9} - \square_{nđ\,9})^2 = (\frac{70}{30} - \frac{862}{10283}.27)^2 = 0{,}0049$

$\Rightarrow \sum_{i=4}^{9}\blacksquare(E_{nđ\,i} - \square_{nđ\,i})^2 = 0{,}0404$

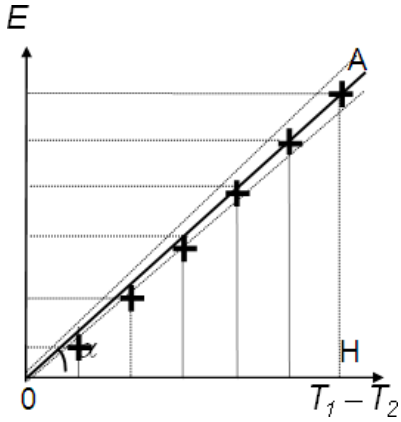
$\square^2 = \frac{1}{9-4}\sum_{i=4}^{9}\blacksquare(E_{nđ\,i} - \square_{nđ\,i})^2 = \frac{1}{5}\sum_{i=4}^{9}\blacksquare(E_{nđ\,i} - \square_{nđ\,i})^2 = \frac{1}{5}.0{,}0404 = 0{,}0081$

- $(\Delta C)^2 = \frac{S^2}{\sum_{i=1}^{n}\blacksquare(T_{1\,i} - T_2)^2}$

$(\Delta C)^2 = \frac{S^2}{\sum_{i=4}^{9}\blacksquare(T_{1\,i} - T_2)^2} = \frac{0{,}0081}{10283} = 7{,}8771.10^{-7}$

$\Delta C = \sqrt{(\Delta C)^2} = \sqrt{7{,}8771.10^{-7}} = 0{,}0009\ (mV/°C)$

3.2 Method 2



a. Calculation of C

Let α be the tilt angle of segment OA with respect to the horizontal axis, we have:

$$\underline{C} = \tan\alpha = \frac{\underline{E}_{\underline{A}}.\underline{T}_{\underline{A}}}{\underline{E}_{\underline{H}}.\underline{T}_{\underline{H}}} = \frac{\frac{127}{30}}{52} = 0{,}0814\ (mV/°C)$$

b. Calculation of error ΔC (Review item IV of the lecture "How to determine the error of the measurement of physical quantities"

Applying differential, we obtain:

$$\Delta C = \Delta(\tan\alpha) = \frac{\Delta\alpha}{\cos^2\alpha} = (1 + \tan^2\alpha).\Delta\alpha$$

Since $C^2 = tg^2\alpha << 1$, we deduce equation 4:

$$\Delta C = (1 + C^2).\Delta\alpha \approx \Delta\alpha \quad (4)$$

Here, we can consider $\Delta\alpha$ the angle between two lines starting from origin O and going through the two endpoints of the error segment 2ΔE. $\Delta\alpha$ is approximated as denoted in equation 5: $\Delta\alpha \approx \frac{2\Delta E}{E}$ (5)

$$E = \sqrt{\underline{E_A}^2 + \underline{E_H}^2} = \sqrt{\left(\frac{127}{30}\right)^2 + 52^2} = 52{,}1720$$

and $2\Delta E = 2.0{,}2513 = 0{,}5026$

Thus, according to equation 4, we obtain: $\Delta C \approx \Delta\alpha \approx \frac{2\Delta E}{E} \approx \frac{0{,}5026}{52{,}1720} = 0{,}0096\ (mV/°C)$

VIII/ CONCLUSION

Method 1: $C = \underline{C} \pm \Delta C = 0{,}0838 \pm 0{,}0009\ (mV/°C)$

Method 2: $C = \underline{C} \pm \Delta C = 0{,}0814 \pm 0{,}0096\ (mV/°C)$

BACHKHOACNCP.COM TÀI LIỆU SƯU TẬP BỞI HCMUT-CNCP